

Motorsport Analytics Platform - Project Proposal

Motorsport Analytics Platform

Project Overview

A web-based motorsport analytics platform that consumes OpenF1 data through a Java Spring Boot backend and presents live and historical Formula 1 insights through a React frontend.

Objectives

- Provide live race and session information.
- Visualize telemetry and performance data.
- Compare drivers, teams, and race strategies.
- Deliver a modern, responsive user experience.
- Create a portfolio-quality full-stack application.

Architecture

OpenF1 API → Spring Boot Backend → REST APIs → React Frontend

Core Features

1. Live Timing Dashboard
2. Driver Comparison Tool
3. Telemetry Visualizer
4. Pit Strategy Analysis
5. Team Radio Hub
6. Race Timeline Viewer
7. Championship Predictor
8. AI-generated Race Insights

Technology Stack

Backend: Java 21, Spring Boot, WebClient, PostgreSQL, Redis

Frontend: React, TypeScript, Axios, React Query, Chart.js/Recharts

Deployment: Docker, GitHub Actions, Cloud Hosting

Roadmap

Phase 1: Core APIs and dashboard

Phase 2: Telemetry and comparisons

Phase 3: Strategy tools and predictions

Phase 4: AI insights and advanced analytics